AIML Transaction and Fraud Analysis – Project Report

1. Introduction

The AIML Transaction and Fraud Analysis dashboard is designed to analyze financial transactions and identify potentially fraudulent activities using Artificial Intelligence and Machine Learning (AI/ML). It presents important transaction details through key performance indicators (KPIs), charts, and interactive filters.

The dashboard helps users understand transaction volumes, transaction amounts, fraudulent transactions, and alert types. It supports financial institutions in monitoring suspicious activities and improving fraud detection.

2. Objectives of the Project

- To analyze a large volume of financial transactions.
- To identify and monitor fraudulent transactions.
- To calculate the total and average transaction amounts.
- To visualize transaction trends over time.
- To analyze fraud alerts based on alert types.
- To support better financial security and decision-making.

3. Analysis of the Dashboard

3.1 Key Performance Indicators (KPIs)

The dashboard displays six major performance indicators.

KPI	Value	Analysis
Total Accounts	10K	The dataset contains approximately 10,000 accounts.
Total Transactions	1M	Approximately one million transactions are represented.

Average Transaction Amount	115.99K	The displayed average transaction value is 115.99K.
Total Transaction Amount	153.48bn	The dashboard reports a total transaction value of 153.48 billion.
Fraudulent Transactions	1,719	The dashboard identifies 1,719 transactions as fraudulent.
Total Initial Balance	2.80M	The reported initial balance is 2.80 million.

These KPIs provide a quick overview of account activity, transaction volume, financial value, and potential fraud.

3.2 Total Transactions by Timestamp

Chart type: Line chart

The line chart represents the number of transactions across timestamps. The transaction count fluctuates over time, with several peaks and declines.

Analysis:

- Transaction activity varies throughout the observed period.
- Peaks indicate timestamps with relatively high transaction activity.
- Sudden changes in transaction volume may warrant further investigation.
- Timestamp analysis can help identify unusual transaction patterns.

3.3 Total Transaction Amount by IS_FRAUD

Chart type: Bar chart

This chart compares transaction amounts according to the `IS_FRAUD` field, which indicates whether a transaction is fraudulent.

Analysis:

- The largest displayed amount is associated with the blank category.
- The bar for `True` is comparatively small.
- The blank category may represent missing or unclassified fraud labels.
- Missing values should be examined before drawing conclusions about fraud losses.

3.4 Total Transactions by IS_FRAUD

Chart type: Doughnut chart

The doughnut chart shows the distribution of transactions by fraud status.

Fraud status	Displayed percentage
False	78.99%
True	21.01%

Analysis:

The chart indicates that the `False` category accounts for approximately 78.99% of the displayed distribution, while the `True` category accounts for 21.01%.

This chart helps compare the relative proportions of transactions classified as fraudulent and non-fraudulent. However, the displayed percentages should be validated against the reported count of 1,719 fraudulent transactions.

3.5 Count of ALERT_ID by ALERT_TYPE

Chart type: Horizontal bar chart

This chart compares the number of alerts for two alert types: `cycle` and `fan_in`.

Analysis:

- The `cycle` alert type has approximately 930 alerts.
- The `fan_in` alert type has approximately 780 alerts.
- The `cycle` alert type is more frequent in the displayed chart.
- Comparing alert types can help prioritize investigations into suspicious transaction patterns.

A cycle alert may relate to transactions forming a loop, while a fan-in pattern may involve funds flowing from multiple sources into one destination. The exact meanings depend on the dataset's alert-generation rules.

3.6 Interactive Filters

The dashboard includes two interactive filtering sections.

IS_FRAUD filter

- False
- True

This filter allows users to focus on fraudulent or non-fraudulent transactions.

ALERT_TYPE filter

- Blank
- fan_in
- cycle

This filter allows users to examine the dashboard for particular alert categories. These filters make it easier to explore specific patterns without reviewing all records individually.

4. Role of AI and Machine Learning

Artificial Intelligence and Machine Learning can improve fraud analysis through the following methods:

1. Anomaly detection: Identifies transactions that differ significantly from normal behavior.
2. Classification: Predicts whether a transaction is likely to be fraudulent based on historical patterns.
3. Pattern recognition: Detects unusual transaction sequences and suspicious account relationships.
4. Risk scoring: Assigns risk scores to transactions to prioritize investigations.
5. Real-time monitoring: Helps flag suspicious transactions as they occur.

The dashboard visualizes transaction and alert data; the image alone does not establish which AI/ML algorithms were implemented or how accurately they detect fraud.

5. Major Findings

The dashboard provides the following observations:

- The dataset contains approximately 10K accounts and 1M transactions.
- The reported fraudulent transaction count is 1,719.
- The displayed total transaction amount is 153.48bn.
- Transaction activity fluctuates across timestamps.
- The `cycle` alert type has a higher displayed count than `fan_in`.
- Interactive filters support focused fraud and alert analysis.

Data validation is important: The dashboard's fraud percentages, fraudulent transaction count, total transaction amount, and average transaction amount may not align with one another. The underlying dataset, aggregation formulas, and treatment of blank values should be checked before the results are used for formal reporting.

6. Advantages of the Project

- Provides a clear visual representation of financial transactions.
- Helps identify potentially suspicious activity.
- Reduces the effort required for manual transaction review.
- Supports data-driven decision-making.
- Enables comparisons between alert categories.
- Provides a foundation for developing automated fraud detection systems.

7. Limitations

- Missing or blank fraud labels may affect the analysis.
- Incorrect data or aggregation formulas may produce misleading results.
- A dashboard cannot independently confirm that a transaction is fraudulent.
- Effective machine learning requires reliable historical data and model evaluation.
- The displayed charts do not provide enough information to determine the accuracy of a fraud detection model.

8. Conclusion

The AIML Transaction and Fraud Analysis project demonstrates how data visualization can be used to monitor financial transactions and analyze potential fraud. The dashboard combines key performance indicators, time-based transaction analysis, fraud-status comparisons, and alert-type analysis in a single interface.

By integrating reliable data processing, machine learning algorithms, and interactive visualizations, this project can support faster identification of suspicious transaction patterns and improve financial risk management. Validating the displayed figures and investigating missing values are essential steps toward obtaining accurate and dependable results.